Confidence Intervals & Hypothesis Testing

ANSWER KEY

Constructing a confidence interval

1 A random sample of 50 students has a mean study time of $\bar{x} = 6.2$ hours per week, with sample standard deviation $s = 1.8$ hours. Construct a 95% confidence interval for the population mean study time, using a critical value of $z^* \approx 1.96$.

Standard error: $\frac{s}{\sqrt{n}} = \frac{1.8}{\sqrt{50}} \approx 0.255$. Margin of error: $1.96(0.255) \approx 0.499$. Confidence interval: 6.2 ± 0.499, or approximately $(5.70, 6.70)$ hours.

2 Interpret the confidence interval from the previous problem in the context of the study time question, using correct statistical language.

We are 95% confident that the true mean weekly study time for the population of all students is between approximately 5.70 and 6.70 hours — meaning that if we repeated this sampling process many times, about 95% of the resulting confidence intervals would capture the true population mean.

Margin of error and sample size

3 Explain two ways to decrease the margin of error of a confidence interval, and state a tradeoff involved with each.

(1) Increase the sample size n: since margin of error is proportional to $\frac{1}{\sqrt{n}}$, a larger sample shrinks the margin of error, but larger samples cost more time and resources to collect. (2) Decrease the confidence level (e.g. from 95% to 90%): a lower confidence level uses a smaller critical value, shrinking the margin of error, but this means being less confident that the interval actually captures the true parameter.

The logic of hypothesis testing

4 State the null hypothesis H_0 and alternative hypothesis H_a for a test of whether a coin is biased toward heads (true proportion of heads p), and classify the test as one-tailed or two-tailed.

$H_0 : p = 0.5$ (the coin is fair). $H_a : p > 0.5$ (the coin is biased toward heads). This is a one-tailed (right-tailed) test, since the alternative specifies a direction.

5 Explain what a p-value represents, in the context of a hypothesis test.

The p-value is the probability of obtaining a test statistic at least as extreme as the one observed, assuming the null hypothesis is true. A small p-value suggests the observed data would be unusual if H_0 were true, providing evidence against H_0.

Four-step hypothesis testing procedure

6 A manufacturer claims their light bulbs last on average $\mu = 1000$ hours. A sample of 40 bulbs has mean lifetime $\bar{x} = 985$ hours with sample standard deviation $s = 45$ hours. State the hypotheses and compute the test statistic for testing whether the true mean lifetime is less than claimed.

$H_0 : \mu = 1000$. $H_a : \mu < 1000$. Test statistic: $z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{985 - 1000}{45/\sqrt{40}} = \frac{-15}{7.115} \approx -2.11$.

7 Using the test statistic $z \approx -2.11$ from the previous problem and a p-value of approximately 0.017, state a conclusion at the $\alpha = 0.05$ significance level.

Since the p-value (0.017) is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence that the true mean bulb lifetime is less than the manufacturer's claimed 1000 hours.

Type I and Type II errors, and power

8 In the light bulb test above, describe what a Type I error would mean in context, and what a Type II error would mean in context.

A Type I error would mean concluding the true mean lifetime is less than 1000 hours when it actually is not (rejecting a true H_0) — wrongly penalizing the manufacturer. A Type II error would mean failing to detect that the true mean lifetime really is less than 1000 hours (failing to reject a false H_0) — missing a real problem with the bulbs.

9 Explain what the power of a test measures, and state one way to increase it.

Power measures the probability that a test correctly rejects H_0 when the alternative hypothesis is actually true (i.e. correctly detecting a real effect). Power can be increased by increasing the sample size, which reduces variability in the sampling distribution and makes real effects easier to detect.

Confidence intervals for a proportion

10 A poll of 400 randomly selected voters finds that 220 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.

$\hat{p} = \frac{220}{400} = 0.55$. Standard error: $\sqrt{\frac{0.55(0.45)}{400}} = \sqrt{\frac{0.2475}{400}} \approx 0.0249$. Margin of error: $1.96(0.0249) \approx 0.0488$. Confidence interval: 0.55 ± 0.0488, or approximately $(0.501, 0.599)$.

Determining sample size for a target margin of error

11 A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a conservative planning value of $p = 0.5$ (which maximizes required sample size). Find the minimum sample size needed, using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with $z^* = 1.96$.

$n = \left(\frac{1.96}{0.03}\right)^2 (0.5)(0.5) = (65.33)^2(0.25) \approx 4268.4(0.25) \approx 1067.1$. Round up: $n = 1068$ (sample size must always be rounded up to guarantee the margin of error requirement is met).

One-sample t-test for a mean

12 A nutritionist claims a new granola bar has a mean of 200 calories. A sample of 15 bars has mean $\bar{x} = 206$ calories with sample standard deviation $s = 9$ calories. Since the population standard deviation is unknown and n is small, a t-test is used. Compute the test statistic $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$.

$t = \frac{206 - 200}{9/\sqrt{15}} = \frac{6}{2.324} \approx 2.58.$

13 Explain what it means for a hypothesis test result to be "statistically significant," and why statistical significance does not always imply practical importance.

A result is statistically significant when the p-value is small enough (below the chosen α) to reject H_0, meaning the observed data would be unlikely if H_0 were true. However, with a very large sample size, even a tiny, practically unimportant difference can become statistically significant — so significance alone doesn't guarantee the effect is large or meaningful in real-world terms; the actual size of the effect should also be considered.

14 A researcher wants to test whether a new fitness program changes the average resting heart rate of participants from the population baseline of $\mu = 72$ bpm. A sample of 36 participants after the program has mean resting heart rate $\bar{x} = 68.5$ bpm with sample standard deviation $s = 10.5$ bpm. State the hypotheses (choosing a two-sided alternative), compute the test statistic, and using a p-value of approximately 0.11, state a conclusion at $\alpha = 0.05$.

$H_0 : \mu = 72$. $H_a : \mu \neq 72$. Test statistic: $t = \frac{68.5 - 72}{10.5/\sqrt{36}} = \frac{-3.5}{1.75} = -2.0$. Since the p-value (0.11) is greater than $\alpha = 0.05$, we fail to reject H_0 — there is not enough evidence that the fitness program changes the average resting heart rate.

15 Explain the relationship between a confidence interval and a two-sided hypothesis test at the same confidence/significance level — specifically, describe how you could use a 95% confidence interval alone to decide whether to reject $H_0 : \mu = \mu_0$ at $\alpha = 0.05$, without separately computing a p-value.

A 95% confidence interval and a two-sided test at $\alpha = 0.05$ are directly linked: if the hypothesized value μ_0 falls outside the 95% confidence interval, then $H_0 : \mu = \mu_0$ would be rejected at $\alpha = 0.05$; if μ_0 falls inside the confidence interval, H_0 would fail to be rejected. This means a confidence interval can be used as a quick visual/numeric check for significance without needing to separately calculate a p-value.

16 A city council is deciding whether to fund a new public transit line, and wants to survey residents to estimate the true proportion who would use it regularly. A preliminary small survey suggests around 35% support, but the council wants a formal survey with a margin of error no larger than 2.5% at 95% confidence. Using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with the preliminary estimate $p = 0.35$ and $z^* = 1.96$, find the required sample size, rounding appropriately.

$n = \left(\frac{1.96}{0.025}\right)^2 (0.35)(0.65) = (78.4)^2 (0.2275) \approx 6146.6(0.2275) \approx 1398.4$. Round up: $n = 1399$ residents (sample size must be rounded up to guarantee the target margin of error is achieved).

17 A restaurant chain wants to test whether a new menu item's average preparation time is longer than the target of 8 minutes, since customers have complained about slow service. A random sample of 30 orders for the new item has a mean preparation time of $\bar{x} = 8.7$ minutes with sample standard deviation $s = 1.8$ minutes. State the hypotheses for this one-sided test, compute the test statistic, and using a p-value of approximately 0.019, state a conclusion at $\alpha = 0.05$ in the context of the restaurant's concern.

$H_0 : \mu = 8$. $H_a : \mu > 8$. Test statistic: $t = \frac{8.7 - 8}{1.8/\sqrt{30}} = \frac{0.7}{0.329} \approx 2.13$. Since the p-value (0.019) is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence that the true mean preparation time for the new menu item exceeds the 8-minute target, supporting the customers' complaints about slow service.